

 [Learn](#)  [Practice](#)

Chemical Bonding & Hybridisation — D4

Issue #38 Candidate Specimen:

Construction

Contents and concept route

Concept Book

Chemical Bonding and Hybridisation

Concept route

1. Conformational torsion and p-orbital alignment
 1. Construct both arrangements while preserving connectivity and electron inventories.
 2. Explain the supplied trend without replacing a mechanism with a label.
 3. Execute the supplied cosine model and separately test the total-energy inference.
2. Delocalisation-sensitive hybridisation boundaries
 1. Resolve conflict between a local Lewis inventory and supplied aligned delocalisation.
 2. Test uniqueness, rather than merely finding one familiar angle.

3. Translate occupancy as well as geometry in the supplied radical model.

4. Separate a discriminating observation from the strength of the inference it warrants.

3. Local domain counting versus conjugated networks

1. Build a decision procedure that can fail its own applicability test.

2. Distinguish recalling a rule from establishing its applicability.

3. Audit a proposed universal rule across cases that stress different boundaries.

Sections

Construct both arrangements while preserving connectivity and electron inventories.

Explain the supplied trend without replacing a mechanism with a label.

Execute the supplied cosine model and separately test the total-energy inference.

Conformational torsion and p-orbital alignment

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Conformational torsion and p-orbital alignment

Concept difficulty: Hard concept

1

Construct both arrangements while preserving connectivity and electron inventories.

Hard concept

Section 1 of 10

Next →

Conditions for this construction

- Understands sigma and pi bonding as head-on and lateral orbital overlaps.
- Can recognize parallel versus perpendicular geometric vectors in 3D.

Why this section matters for the question set

This unit builds toward

- Q2 · D4

The move learners miss: Construct the supplied aligned donor/carbonyl pi account. [Try it in Core2](#)

1 Inventory formamide electrons and connectivity.

Why valid: CH3NO supplies 4+3+5+6 valence electrons and 6+3+7+8 total electrons.

Result: 18 valence; 24 total; H-C(=O)-NH2 connectivity fixed.

2 The step the hard question turns on Construct the supplied aligned donor/carbonyl pi account.

Why valid: The localized donor pair plus carbonyl pi pair gives four electrons in a three-centre qualitative pi space.

Result: Aligned N p-like donor and C/O p directions; four model pi-space electrons.

3 Rotate the N donor direction about C-N toward 90 degrees.

Why valid: Relative torsion changes orientation against the fixed carbonyl reference; it does not change element counts.

Result: Approximately orthogonal donor/acceptor directions with unchanged connectivity and electron ledger.

4 Check what the two sketches can establish.

Why valid: Orientation distinguishes alignment, while donation strength needs an experiment or specified calculation.

Result: Both drawings retain 18 valence/24 total electrons; no measured donation strength is claimed.

Fixed connectivity: H-C(=O End-on view along C-N; ce

Connectivity in the declared qualitative direction model.

Stage 1 of 5

Construct local directions in an explicitly declared end-on projection; preserve connectivity, occupancy and model limits.

Keep alignment separate from total energy

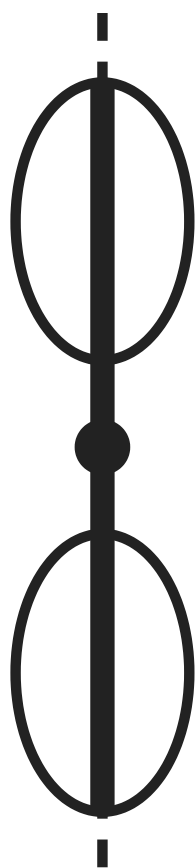
Look along the fixed C-N axis. The two centres project to the same origin. The solid donor direction turns relative to the dashed carbonyl direction. This is a direction diagram, not a measured orbital or overlap integral.

Relative angle θ (degrees)



Independent toy energy contribution c





Axis points towards/away from viewer.

$\theta = 0^\circ$; given factor = 1; toy $E = -1$.

Declared demonstration only: $g = \cos(\theta)$; $E = -g^2 + c$ in arbitrary toy units. The square is an explicitly chosen example, not a universal law. Hold θ fixed and vary c : the same g permits different E . Predict this before moving c . No physical barrier is computed.

The key step

Twisting reduces relevant alignment and the supplied delocalisation contribution. A changing hybrid label alone explains neither mechanism nor a quantitative barrier; total torsional energy has additional contributions.

Equations and validity

Equation	What it tells you	When you can
Given direction factor $g(\theta) = \cos(\theta)$; total energy unspecified.	Given factors at 0, 60, 90 degrees are 1, 1/2, 0. Both cosine and cosine-squared are even. A degenerate two-orbital Hamiltonian with coupling $t_0 \cos(\theta)$ gives $\epsilon \pm t_0 \cos(\theta) $, so no universal cosine-squared total-energy law follows.	- Applies to overlap between two p orbitals rotated at their common sigma internuclear axis. - At $\theta = 0$ by symmetry, orbitals are orthogonal.
Check an ordinary contiguous appropriately aligned p-bearing path; this is a bounded model criterion.	Amide A has the supplied direct donor/carbonyl path. Saturated-spacer amine B lacks that ordinary contiguous path. Spatial proximity alone is insufficient, and absence of this path is not an absolute zero-interaction theorem.	- All atoms conjugated network must have unhybridized orbitals. - Adjacent orbitals must have compatible symmetry alignment.

Worked explanation

Count valence and total electrons for CH3NO before and after torsion. Identify which geometric relation changes.

Follow the operation, its justification and its result.

- Torsion changes relative orientation, not the number of atoms or formal electrons.

2. 18 valence, 24 total in both; relative donor direction changes.
3. Formamide CH_3NO has 18 valence electrons and 24 total electrons. The supplied three-centre donor/carbonyl π model has four electrons. A 90-degree twist changes alignment, not these inventories.

Result

18 valence, 24 total in both; relative donor direction changes.

Check

Independent element sums: valence $4+3+5+6=18$; total $6+3+7+8=24$. Both sketches must preserve them.

Mistake clinic • diagnose and repair

A tempting wrong path

- Torsion changes the formal electron inventory.
- A changed hybrid label is a causal explanation of donor strength.
- The given overlap factor is the total molecular energy.

Diagnose

- Count valence and total electrons for CH_3NO before and after torsion. Identify which geometric relation changes.
- If two sketches have the same label but different donor alignment, must their donation be identical? If the label changes alone, has a mechanism been supplied?
- If θ stays at 60 degrees while an independent energy term changes, does overlap change? Does total energy necessarily stay fixed?

Repair

- Formamide CH_3NO has 18 valence electrons and 24 total electrons. The supplied three-centre donor/carbonyl π model has four electrons. A 90-degree twist changes alignment, not these inventories.
- Twisting reduces relevant alignment and the supplied delocalisation contribution. A changing hybrid label alone explains neither mechanism

nor a quantitative barrier; total torsional energy has additional contributions.

- Given factors at 0,60,90 degrees are 1,1/2,0. Both cosine and cosine-squared are even. A degenerate two-orbital Hamiltonian with coupling to $\cos(\theta)$ gives $\epsilon \pm |t_0 \cos(\theta)|$, so no universal cosine-squared total-energy law follows.

1-2-3 QUICK CHECK

1 • CHECK

Independent element sums: valence $4+3+5+6=18$; total $6+3+7+8=24$. Both sketches must preserve them.

2 • APPLY

Sketch a fixed-connectivity 45-degree comparison without claiming measured donation.

3 • CONNECT

Torsion changes relative orientation, not the number of atoms or formal electrons.

2

Explain the supplied trend without replacing a mechanism with a label.

Hard concept

Section 2 of 10

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- Q3 · D4

The move learners miss: Connect torsion to the overlap mechanism.

[Try it in Core2](#)

1 Use the supplied model result as a conditional premise.

Why valid: The benchmark specifies reduced alignment and delocalisation contribution toward 90-degree twist.

Result: Connectivity fixed; relevant orbital directions rotate relative to each other.

2 The step the hard question turns on Connect torsion to the overlap mechanism.

Why valid: Parallel p-like donor/acceptor directions support lateral interaction; orthogonal directions reduce that particular alignment.

Result: Reduced alignment explains the stated qualitative contribution change.

3 Test the hybrid-label-only explanation.

Why valid: A label does not encode the adjacent acceptor or their relative orientation.

Result: Renaming hybridisation alone is insufficient; the alignment pathway supplies the mechanism.

4 Separate model contribution from measured total energy.

Why valid: Other energy terms and relaxation can also vary under torsion.

Result: No unique physical basis, numerical barrier or full total-energy law follows from this qualitative account.

Fixed connectivity: H-C(=O End-on view along C-N; ce

Connectivity in the declared qualitative direction model.

Stage 1 of 5

Construct local directions in an explicitly declared end-on projection; preserve connectivity, occupancy and model limits.

Worked explanation

If two sketches have the same label but different donor alignment, must their donation be identical? If the label changes alone, has a mechanism been supplied?

Follow the operation, its justification and its result.

1. Relative donor/acceptor alignment is a model variable; donation magnitude needs additional physics.
2. Neither follows.
3. Twisting reduces relevant alignment and the supplied delocalisation contribution. A changing hybrid label alone explains neither mechanism nor a quantitative barrier; total torsional energy has additional contributions.

Result

Neither follows.

Check

Counterfactual: remove the word hybridisation while retaining changed orientation; the mechanistic statement survives.

Mistake clinic • diagnose and repair

A tempting wrong path

- Torsion changes the formal electron inventory.
- A changed hybrid label is a causal explanation of donor strength.
- The given overlap factor is the total molecular energy.

Diagnose

- Count valence and total electrons for CH₃NO before and after torsion. Identify which geometric relation changes.
- If two sketches have the same label but different donor alignment, must their donation be identical? If the label changes alone, has a mechanism been supplied?

- If theta stays at 60 degrees while an independent energy term changes, does overlap change? Does total energy necessarily stay fixed?

Repair

- Formamide CH_3NO has 18 valence electrons and 24 total electrons. The supplied three-centre donor/carbonyl pi model has four electrons. A 90-degree twist changes alignment, not these inventories.
- Twisting reduces relevant alignment and the supplied delocalisation contribution. A changing hybrid label alone explains neither mechanism nor a quantitative barrier; total torsional energy has additional contributions.
- Given factors at 0,60,90 degrees are 1,1/2,0. Both cosine and cosine-squared are even. A degenerate two-orbital Hamiltonian with coupling $t_0 \cos(\theta)$ gives $\epsilon \pm |t_0 \cos(\theta)|$, so no universal cosine-squared total-energy law follows.

1-2-3 QUICK CHECK

1 • CHECK

Counterfactual: remove the word hybridisation while retaining changed orientation; the mechanistic statement survives.

2 • APPLY

State which additional model would be needed for a numerical donation or barrier estimate.

3 • CONNECT

Relative donor/acceptor alignment is a model variable; donation magnitude needs additional physics.

3

Execute the supplied cosine model and separately test the total-energy inference.

Hard concept

Section 3 of 10

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- **Q9 · D4**

The move learners miss: Identify the missing link from overlap to total energy. [Try it in Core2](#)

1 Evaluate the supplied overlap factor at the three angles.

Why valid: The stated model is $\cos(\theta)$; $\cos(0)=1$, $\cos(60)=1/2$ and $\cos(90)=0$.

Result: Factors: 1, 1/2, 0.

2 **The step the hard question turns on** Identify the missing link from overlap to total energy.

Why valid: An overlap factor is not a Hamiltonian and supplies no full energy budget.

Result: The proposed total-energy conclusion is not warranted.

3 Construct an explicit counterexample to uniqueness.

Why valid: At fixed $\theta=60$ an independently specified energy contribution can vary while the overlap factor stays 1/2.

Result: Same overlap, different total energy; proportionality is not implied.

4 Bound even alternative energy approximations.

Why valid: A squared-interaction estimate needs stated weak-coupling assumptions; a degenerate two-level coupling can instead yield ϵ plus/minus absolute t .

Result: No universal cosine or cosine-squared total-energy law; both functions are even, so parity does not select one.

Fixed connectivity: H-C(=O

End-on view along C-N; ce

Connectivity in the declared qualitative direction model.

Stage 1 of 5

Construct local directions in an explicitly declared end-on projection; preserve connectivity, occupancy and model limits.

Keep alignment separate from total energy

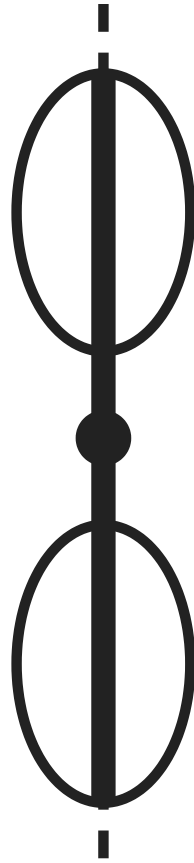
Look along the fixed C-N axis. The two centres project to the same origin. The solid donor direction turns relative to the dashed carbonyl direction. This is a direction diagram, not a measured orbital or overlap integral.

Relative angle θ (degrees)



Independent toy energy contribution c





Axis points towards/away from viewer.

$\theta = 0^\circ$; given factor = 1; toy $E = -1$.

Declared demonstration only: $g = \cos(\theta)$; $E = -g^2 + c$ in arbitrary toy units. The square is an explicitly chosen example, not a universal law. Hold θ fixed and vary c : the same g permits different E . Predict this before moving c . No physical barrier is computed.

Worked explanation

If theta stays at 60 degrees while an independent energy term changes, does overlap change? Does total energy necessarily stay fixed?

Follow the operation, its justification and its result.

1. An overlap factor is not an energy Hamiltonian or the whole energy budget.

2. Overlap factor unchanged; total energy need not remain fixed.
3. Given factors at 0,60,90 degrees are 1,1/2,0. Both cosine and cosine-squared are even. A degenerate two-orbital Hamiltonian with coupling $t_0 \cos(\theta)$ gives $\epsilon \pm |t_0 \cos(\theta)|$, so no universal cosine-squared total-energy law follows.

Result

Overlap factor unchanged; total energy need not remain fixed.

Check

At $\theta=60$ degrees the factor is fixed at 1/2 while adding an independently specified energy contribution changes total energy. This falsifies uniqueness without modelling formamide quantitatively.

Mistake clinic · diagnose and repair

A tempting wrong path

- Torsion changes the formal electron inventory.
- A changed hybrid label is a causal explanation of donor strength.
- The given overlap factor is the total molecular energy.

Diagnose

- Count valence and total electrons for CH3NO before and after torsion. Identify which geometric relation changes.
- If two sketches have the same label but different donor alignment, must their donation be identical? If the label changes alone, has a mechanism been supplied?
- If θ stays at 60 degrees while an independent energy term changes, does overlap change? Does total energy necessarily stay fixed?

Repair

- Formamide CH3NO has 18 valence electrons and 24 total electrons. The supplied three-centre donor/carbonyl pi model has four electrons. A 90-degree twist changes alignment, not these inventories.

- Twisting reduces relevant alignment and the supplied delocalisation contribution. A changing hybrid label alone explains neither mechanism nor a quantitative barrier; total torsional energy has additional contributions.
- Given factors at 0,60,90 degrees are 1,1/2,0. Both cosine and cosine-squared are even. A degenerate two-orbital Hamiltonian with coupling $t_0 \cos(\theta)$ gives $\epsilon \pm |t_0 \cos(\theta)|$, so no universal cosine-squared total-energy law follows.

1-2-3 QUICK CHECK

1 • CHECK

At $\theta=60$ degrees the factor is fixed at 1/2 while adding an independently specified energy contribution changes total energy. This falsifies uniqueness without modelling formamide quantitatively.

2 • APPLY

State assumptions needed for a weak-coupling second-order squared-interaction estimate.

3 • CONNECT

An overlap factor is not an energy Hamiltonian or the whole energy budget.

Question-specific diagnosis and repair

Repair for Q2

Test the rule on a changed case

Count valence and total electrons for CH3NO before and after torsion.

Identify which geometric relation changes.

Your prediction and reason

Repair for Q3

Test the rule on a changed case

If two sketches have the same label but different donor alignment, must their donation be identical? If the label changes alone, has a mechanism been supplied?

Your prediction and reason

Repair for Q9

Test the rule on a changed case

If theta stays at 60 degrees while an independent energy term changes, does overlap change? Does total energy necessarily stay fixed?

Your prediction and reason

Now you do one

For a supplied direction factor $g(\theta) = \cos(\theta)$, compare 30° and 60° . Does the factor alone specify a total-energy difference? Hold 60° fixed and add a toy energy contribution c .

Model answer

The factors are $\sqrt{3}/2$ and $1/2$. No total-energy difference follows without an energy model. In the explicitly chosen toy family $E = -g^2 + c$, varying c at fixed 60° changes E while g stays $1/2$.

1. The factors are $\sqrt{3}/2$ and $1/2$. No total-energy difference follows without an energy model. In the explicitly chosen toy family $E = -g^2 + c$, varying c at fixed 60° changes E while g stays $1/2$.

Practice this concept in Core2

Q2

Q3

Q9

Sections

Resolve conflict between a local Lewis inventory and supplied aligned delocalisation.

Test uniqueness, rather than merely finding one familiar angle.

Translate occupancy as well as geometry in the supplied radical model.

Separate a discriminating observation from the strength of the inference it warrants.

Delocalisation-sensitive hybridisation boundaries

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Delocalisation-sensitive hybridisation boundaries

Concept difficulty: Hard concept

1

Resolve conflict between a local Lewis inventory and supplied aligned delocalisation.

Hard concept

Section 4 of 10

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Next →

Conditions for this construction

- Can determine local Lewis structures and count electron domains.

- Understands resonance stabilization in introductory conjugated systems.

Why this section matters for the question set

This unit builds toward

- **Q1** · D4 · the hardest question in this set
The move learners miss: Choose a basis that represents both jobs.
[Try it in Core2](#)

1 Record the Lewis inventory and the supplied donation condition separately.

Why valid: N has three bonds plus a lone pair, but the task also supplies approximately planar donation into an adjacent carbonyl pi system.

Result: A four-entry Lewis ledger does not settle the electronic model.

2 The step the hard question turns on Choose a basis that represents both jobs.

Why valid: Three in-plane sp²-like sigma directions can coexist with an aligned p-like lone-pair donor under the supplied model.

Result: A planar sigma framework plus a p-like donor connected to the carbonyl pi space.

3 Explain why automatic sp³ assignment is inadequate.

Why valid: A localized tetrahedral rule omits the stated delocalisation and alignment conditions.

Result: The local count is a useful ledger, not an overriding proof of the basis.

4 Bound and counterfactually test the description.

Why valid: Twisting or interrupting the donor-acceptor path changes the applicability; geometry alone does not uniquely measure hybrid coefficients.

Result: An introductory sp²-like account is conditional, not unique observed orbital reality.

Fixed connectivity: H-C(=O

End-on view along C-N; ce

Connectivity in the declared qualitative direction model.

Stage 1 of 5

Construct local directions in an explicitly declared end-on projection; preserve connectivity, occupancy and model limits.

The key step

Approximately planar amide N can use an introductory sp²-like sigma framework plus an aligned p-like lone-pair donor. Four Lewis entries do not override the supplied delocalised-model conditions; the basis is not unique observed reality.

Formula reference • meaning and conditions

Equations and validity

Equation	What it tells you	When you can us
$n \text{ \texttt{atomic orbitals}} \rightarrow n \text{ \texttt{hybrid/molecular orbitals}}$	Approximately planar amide N can use an introductory sp ² -like sigma framework plus an aligned p-like lone-pair donor. Four Lewis entries do not override the supplied delocalised-model conditions; the basis is not unique observed reality.	• Conservation orbital dimensionality under unitary transformation • Valence bond hybridisation creates direct equivalent orbitals for sigma bonds while leaving

Equation	What it tells you	When you can use
		remaining p for pi delocalization
Check an ordinary contiguous appropriately aligned p-bearing path; this is a bounded model criterion.	Amide A has the supplied direct donor/carbonyl path. Saturated-spacer amine B lacks that ordinary contiguous path. Spatial proximity alone is insufficient, and absence of this path is not an absolute zero-interaction theorem.	- All atoms in the conjugated system must have unhybridised orbitals. - Adjacent p orbitals must have constructive symmetry alignment.

Worked explanation

Compare adjacent amide N with a saturated-spacer amine N. Does the same four-entry Lewis inventory establish the same donation pathway?

Follow the operation, its justification and its result.

1. A hybrid label describes a chosen basis; it does not independently cause donation.
2. No. Adjacency/alignment conditions differ. Repeating automatic four-domains-therefore-localized-sp³ in both exposes applicability failure.
3. Approximately planar amide N can use an introductory sp²-like sigma framework plus an aligned p-like lone-pair donor. Four Lewis entries do not override the supplied delocalised-model conditions; the basis is not unique observed reality.

Result

No. Adjacency/alignment conditions differ. Repeating automatic four-domains-therefore-localized-sp³ in both exposes applicability failure.

Check

Remove the hybridisation label: the alignment-plus-acceptor explanation must still describe the supplied donation condition.

Mistake clinic • diagnose and repair

—

A tempting wrong path

- Four Lewis entries overrule supplied delocalisation conditions.
- A hybrid label uniquely determines every angle and physical orbital basis.
- Three sigma bonds always imply an empty residual p orbital.
- Geometry alone uniquely measures donation and a rotational barrier.

Diagnose

- Compare adjacent amide N with a saturated-spacer amine N. Does the same four-entry Lewis inventory establish the same donation pathway?
- Can one four-domain label predict the same nuclear shape for CH₄ and H₂O? Does observing a shape uniquely establish an orbital basis?
- Compare CH₃ radical (7 valence electrons) with a model CH₃⁺ inventory (6). Must their residual p occupancy be the same?
- Does observed planar geometry alone measure donor strength? What extra information would a numerical rotational barrier require?

Repair

- Approximately planar amide N can use an introductory sp²-like sigma framework plus an aligned p-like lone-pair donor. Four Lewis entries do not override the supplied delocalised-model conditions; the basis is not unique observed reality.
- CH₄/NH₃/H₂O share common sp³-like introductory descriptions while their nuclear shapes/angles differ. Geometry constrains models but does not uniquely measure a hybrid decomposition; lone pairs and delocalisation matter.
- Supplied planar CH₃ radical has three sigma directions and a perpendicular p-like SOMO with one electron. Three sigma bonds alone do not determine an empty residual p orbital.
- Planarity/bond metrics and condition-specific spectroscopic/dynamic evidence can test localized versus delocalised accounts. Geometry alone

does not uniquely measure donation, hybrid coefficients or a rotational barrier; exact empirical numbers need species/conditions/source.

1-2-3 QUICK CHECK

1 • CHECK

Remove the hybridisation label: the alignment-plus-acceptor explanation must still describe the supplied donation condition.

2 • APPLY

Give one observation that constrains the model and one claim it cannot uniquely prove.

3 • CONNECT

A hybrid label describes a chosen basis; it does not independently cause donation.

2

Test uniqueness, rather than merely finding one familiar angle.

Hard concept

Section 5 of 10

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Next →

Why this section matters for the question set

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This unit builds toward

- Q5 · D4

The move learners miss: Reject the unique-angle implication.

[Try it in Core2](#)

1 Compare the supplied angles with their shared introductory label.

Why valid: CH₄, NH₃ and H₂O use sp³-like descriptions but have different lone-pair counts and nuclear shapes.

Result: 109.5, about 107 and about 104.5 degrees are compatible with the supplied shared label.

2 The step the hard question turns on Reject the unique-angle implication.

Why valid: A single model label does not fix all interactions that affect geometry.

Result: Angle deviations alone do not falsify every sp³-like description.

3 Include the amide boundary in the replacement rule.

Why valid: An aligned adjacent acceptor and delocalisation can make automatic localized four-domain classification inadequate.

Result: Local counts guide a bounded starting model; lone pairs, alignment and delocalisation require additional checks.

4 Specify evidence needed for stronger physical claims.

Why valid: Nuclear geometry alone does not uniquely recover wavefunctions or a hybrid decomposition.

Result: Use defined spectroscopic/electronic-structure evidence and model assumptions; retain alternative compatible descriptions.

Fixed connectivity: H-C(=O End-on view along C-N; ce

Connectivity in the declared qualitative direction model.

Stage 1 of 5

Construct local directions in an explicitly declared end-on projection; preserve connectivity, occupancy and model limits.

Worked explanation

Can one four-domain label predict the same nuclear shape for CH₄ and H₂O?
Does observing a shape uniquely establish an orbital basis?

Follow the operation, its justification and its result.

1. Several shapes/angles can share an introductory four-domain classification.
2. No to both.
3. CH₄/NH₃/H₂O share common sp³-like introductory descriptions while their nuclear shapes/angles differ. Geometry constrains models but does not uniquely measure a hybrid decomposition; lone pairs and delocalisation matter.

Result

No to both.

Check

Independent counterexample: a shared four-domain label is compatible with 4,3,2 bonded-atom positions, so it cannot alone uniquely specify atom-only shape.

Mistake clinic · diagnose and repair

—

A tempting wrong path

- Four Lewis entries overrule supplied delocalisation conditions.
- A hybrid label uniquely determines every angle and physical orbital basis.
- Three sigma bonds always imply an empty residual p orbital.
- Geometry alone uniquely measures donation and a rotational barrier.

Diagnose

- Compare adjacent amide N with a saturated-spacer amine N. Does the same four-entry Lewis inventory establish the same donation pathway?
- Can one four-domain label predict the same nuclear shape for CH₄ and H₂O? Does observing a shape uniquely establish an orbital basis?
- Compare CH₃ radical (7 valence electrons) with a model CH₃⁺ inventory (6). Must their residual p occupancy be the same?
- Does observed planar geometry alone measure donor strength? What extra information would a numerical rotational barrier require?

Repair

- Approximately planar amide N can use an introductory sp^2 -like sigma framework plus an aligned p-like lone-pair donor. Four Lewis entries do not override the supplied delocalised-model conditions; the basis is not unique observed reality.
- $CH_4/NH_3/H_2O$ share common sp^3 -like introductory descriptions while their nuclear shapes/angles differ. Geometry constrains models but does not uniquely measure a hybrid decomposition; lone pairs and delocalisation matter.
- Supplied planar CH_3 radical has three sigma directions and a perpendicular p-like SOMO with one electron. Three sigma bonds alone do not determine an empty residual p orbital.
- Planarity/bond metrics and condition-specific spectroscopic/dynamic evidence can test localized versus delocalised accounts. Geometry alone does not uniquely measure donation, hybrid coefficients or a rotational barrier; exact empirical numbers need species/conditions/source.

1-2-3 QUICK CHECK

1 • CHECK

Independent counterexample: a shared four-domain label is compatible with 4,3,2 bonded-atom positions, so it cannot alone uniquely specify atom-only shape.

2 • APPLY

Name evidence needed before claiming a unique physical electronic description.

3 • CONNECT

Several shapes/angles can share an introductory four-domain classification.

3

Translate occupancy as well as geometry in the supplied radical model.

Hard concept

Section 6 of 10

← Previous

Next →

Why this section matters for the question set

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This unit builds toward

- **Q7 · D3**

The move learners miss: Allocate the carbon valence basis. [Try it in Core2](#)

1 Read the supplied radical geometry and occupancy.

Why valid: The benchmark specifies three planar C-H sigma directions and a perpendicular singly occupied p-like orbital.

Result: A three-direction sigma frame plus one unpaired electron.

2 The step the hard question turns on Allocate the carbon valence basis.

Why valid: An sp²-like sigma allocation uses three functions and leaves one p-like function perpendicular.

Result: Three hybrids plus one residual p-like function; four functions total.

3 Place the supplied electron in the residual orbital.

Why valid: CH₃ radical has 4+3=7 valence electrons; three sigma pairs use six.

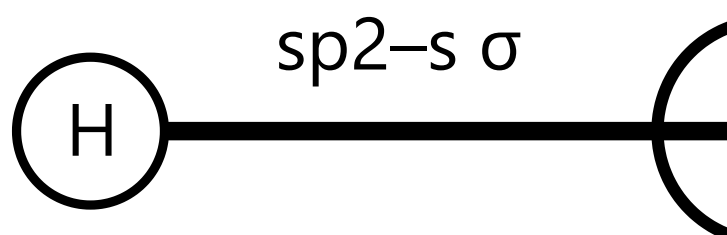
Result: Residual p-like SOMO contains one electron, not zero.

4 Compare with a six-electron CH₃⁺ model inventory.

Why valid: Removing one electron changes residual occupancy while the three-bond count can remain.

Result: Three sigma bonds alone do not determine whether residual p is empty; the given radical model is not a universal claim.

1 · Planar sp^2 carbon sigma framework



Shows three C–H sigma bonds at 120° in the xy plane.

Stage 1 of 3

Staged orbital model of neutral methyl radical $\text{CH}_3^\bullet$, showing the trigonal planar sp^2 sigma framework and the perpendicular unhybridised p_z orbital holding one unpaired electron (SOMO).

Worked explanation

Compare CH_3 radical (7 valence electrons) with a model CH_3^+ inventory (6). Must their residual p occupancy be the same?

Follow the operation, its justification and its result.

1. A singly occupied orbital is not an empty orbital or a lone pair.

2. No: one versus zero in these supplied local models. Repeating empty p for both reveals bonds-only occupancy inference.
3. Supplied planar CH₃ radical has three sigma directions and a perpendicular p-like SOMO with one electron. Three sigma bonds alone do not determine an empty residual p orbital.

Result

No: one versus zero in these supplied local models. Repeating empty p for both reveals bonds-only occupancy inference.

Check

Independent count: CH₃ radical has $4+3=7$ valence electrons; three sigma pairs use six, leaving one.

Mistake clinic • diagnose and repair

A tempting wrong path

- Four Lewis entries overrule supplied delocalisation conditions.
- A hybrid label uniquely determines every angle and physical orbital basis.
- Three sigma bonds always imply an empty residual p orbital.
- Geometry alone uniquely measures donation and a rotational barrier.

Diagnose

- Compare adjacent amide N with a saturated-spacer amine N. Does the same four-entry Lewis inventory establish the same donation pathway?
- Can one four-domain label predict the same nuclear shape for CH₄ and H₂O? Does observing a shape uniquely establish an orbital basis?
- Compare CH₃ radical (7 valence electrons) with a model CH₃⁺ inventory (6). Must their residual p occupancy be the same?
- Does observed planar geometry alone measure donor strength? What extra information would a numerical rotational barrier require?

Repair

- Approximately planar amide N can use an introductory sp²-like sigma framework plus an aligned p-like lone-pair donor. Four Lewis entries do

not override the supplied delocalised-model conditions; the basis is not unique observed reality.

- CH₄/NH₃/H₂O share common sp³-like introductory descriptions while their nuclear shapes/angles differ. Geometry constrains models but does not uniquely measure a hybrid decomposition; lone pairs and delocalisation matter.
- Supplied planar CH₃ radical has three sigma directions and a perpendicular p-like SOMO with one electron. Three sigma bonds alone do not determine an empty residual p orbital.
- Planarity/bond metrics and condition-specific spectroscopic/dynamic evidence can test localized versus delocalised accounts. Geometry alone does not uniquely measure donation, hybrid coefficients or a rotational barrier; exact empirical numbers need species/conditions/source.

1-2-3 QUICK CHECK

1 • CHECK

Independent count: CH₃ radical has 4+3=7 valence electrons; three sigma pairs use six, leaving one.

2 • APPLY

Sketch the given radical while explicitly marking the singly occupied orbital.

3 • CONNECT

A singly occupied orbital is not an empty orbital or a lone pair.

4

Separate a discriminating observation from the strength of the inference it warrants.

Hard concept

Section 7 of 10

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- Q8 · D4

The move learners miss: Propose controlled geometry and bond measurements. [Try it in Core2](#)

1 State competing predictions before proposing observations.

Why valid: The proposed localized pyramidal and donation-compatible planar accounts differ in nuclear arrangement and possible electronic consequences.

Result: Compare N geometry, relevant bond metrics and condition-specific electronic or dynamic signals.

2 The step the hard question turns on Propose controlled geometry and bond measurements.

Why valid: Defined species, phase and conditions allow predictions to be compared with uncertainty and alternative causes.

Result: Near-planarity can challenge a strictly pyramidal description; bond metrics add constraints.

3 Add an independent electronic or torsional test.

Why valid: A specified spectrum or calculation/experiment can test donation-related or rotation-related predictions under stated assumptions.

Result: A second channel of evidence can discriminate accounts more strongly than geometry alone.

4 Limit the warranted inference.

Why valid: Compatible geometry can arise under multiple electronic descriptions; strength and barrier require quantitative evidence.

Result: No unique hybrid coefficients, donation magnitude or rotational barrier is measured by a qualitative planar sketch.

Fixed connectivity: H-C(=O
End-on view along C-N; ce

Connectivity in the declared qualitative direction model.

Stage 1 of 5

Construct local directions in an explicitly declared end-on projection; preserve connectivity, occupancy and model limits.

Worked explanation

Does observed planar geometry alone measure donor strength? What extra information would a numerical rotational barrier require?

Follow the operation, its justification and its result.

1. A measurement can constrain multiple models without uniquely selecting an orbital decomposition.
2. No; a specified experiment or calculation and conditions/model are needed. Repeating direct geometry-to-strength inference exposes overinterpretation.
3. Planarity/bond metrics and condition-specific spectroscopic/dynamic evidence can test localized versus delocalised accounts. Geometry alone does not uniquely measure donation, hybrid coefficients or a rotational barrier; exact empirical numbers need species/conditions/source.

Result

No; a specified experiment or calculation and conditions/model are needed. Repeating direct geometry-to-strength inference exposes overinterpretation.

Check

Use two compatible electronic descriptions of a planar structure as a counterexample to geometry-implies-unique-orbitals.

Mistake clinic • diagnose and repair

A tempting wrong path

- Four Lewis entries overrule supplied delocalisation conditions.
- A hybrid label uniquely determines every angle and physical orbital basis.
- Three sigma bonds always imply an empty residual p orbital.
- Geometry alone uniquely measures donation and a rotational barrier.

Diagnose

- Compare adjacent amide N with a saturated-spacer amine N. Does the same four-entry Lewis inventory establish the same donation pathway?
- Can one four-domain label predict the same nuclear shape for CH₄ and H₂O? Does observing a shape uniquely establish an orbital basis?
- Compare CH₃ radical (7 valence electrons) with a model CH₃⁺ inventory (6). Must their residual p occupancy be the same?
- Does observed planar geometry alone measure donor strength? What extra information would a numerical rotational barrier require?

Repair

- Approximately planar amide N can use an introductory sp²-like sigma framework plus an aligned p-like lone-pair donor. Four Lewis entries do not override the supplied delocalised-model conditions; the basis is not unique observed reality.
- CH₄/NH₃/H₂O share common sp³-like introductory descriptions while their nuclear shapes/angles differ. Geometry constrains models but does not uniquely measure a hybrid decomposition; lone pairs and delocalisation matter.
- Supplied planar CH₃ radical has three sigma directions and a perpendicular p-like SOMO with one electron. Three sigma bonds alone do not determine an empty residual p orbital.
- Planarity/bond metrics and condition-specific spectroscopic/dynamic evidence can test localized versus delocalised accounts. Geometry alone does not uniquely measure donation, hybrid coefficients or a rotational barrier; exact empirical numbers need species/conditions/source.

1-2-3 QUICK CHECK

1 • CHECK

Use two compatible electronic descriptions of a planar structure as a counterexample to geometry-implies-unique-orbitals.

2 • APPLY

Design a second independent observation and state one alternative explanation.

3 • CONNECT

A measurement can constrain multiple models without uniquely selecting an orbital decomposition.

Question-specific diagnosis and repair

Repair for Q1

Test the rule on a changed case

Compare adjacent amide N with a saturated-spacer amine N. Does the same four-entry Lewis inventory establish the same donation pathway?

Your prediction and reason

Repair for Q5

Test the rule on a changed case

Can one four-domain label predict the same nuclear shape for CH₄ and H₂O? Does observing a shape uniquely establish an orbital basis?

Your prediction and reason

Repair for Q7

Test the rule on a changed case

Compare CH₃ radical (7 valence electrons) with a model CH₃⁺ inventory (6). Must their residual p occupancy be the same?

Your prediction and reason

Repair for Q8

Test the rule on a changed case

Does observed planar geometry alone measure donor strength? What extra information would a numerical rotational barrier require?

Your prediction and reason

Now you do one

A hypothetical p-like donor is supplied as aligned with a directly adjacent pi acceptor. Rotate its direction about the fixed connecting bond. What qualitative conclusion follows in this model, and what physical claim still needs evidence?

Model answer

Relative alignment can decrease without changing the formal electron inventory. A quantitative donation strength or total barrier requires a specified physical model or experiment.

1. Relative alignment can decrease without changing the formal electron inventory. A quantitative donation strength or total barrier requires a specified physical model or experiment.

Practice this concept in Core2

Q1

Q5

Q7

Q8

Sections

Build a decision procedure that can fail its own applicability test.

Distinguish recalling a rule from establishing its applicability.

Audit a proposed universal rule across cases that stress different boundaries.

Local domain counting versus conjugated networks

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Local domain counting versus conjugated networks

Concept difficulty: Hard concept

1

Build a decision procedure that can fail its own applicability test.

Hard concept

Section 8 of 10

← Previous

Next →

Conditions for this construction

- Understands orbital hybridisation in alkanes and alkenes.

Why this section matters for the question set

This unit builds toward

- Q4 · D4

The move learners miss: Trace donor-to-carbonyl adjacency and p availability. [Try it in Core2](#)

1 Apply the same N Lewis ledger at both sites.

Why valid: Both neutral N sites have three sigma bonds and one lone pair.

Result: Local count alone does not distinguish A from B.

2 The step the hard question turns on Trace donor-to-carbonyl adjacency and p availability.

Why valid: A has direct N-C(=O) adjacency; B contains the supplied saturated CH₂ interruption.

Result: A supports the stated contiguous donor/acceptor path; B lacks that ordinary path.

3 Test alignment and validity before applying a label.

Why valid: Donation requires suitable orbital orientation as well as adjacency; spatial proximity does not create conjugation.

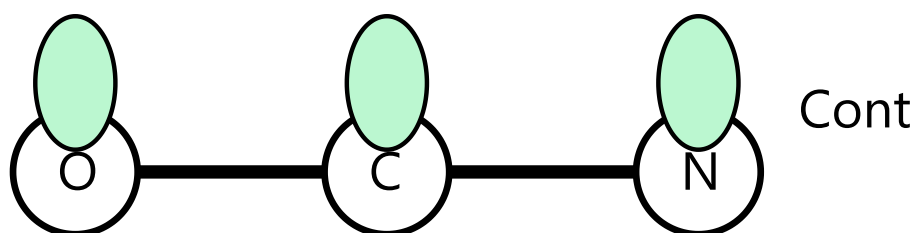
Result: Use conditional delocalised accounting at A and localized amine accounting at B within the given model.

4 Construct a structural counterfactual.

Why valid: Replacing the saturated interruption with a p-bearing aligned bridge changes the path test.

Result: Reassess the model; absence of this ordinary path is not an absolute zero-interaction theorem.

Site A · Direct amide: continuous aligned p-orbitals



Shows adjacent sp² centers with aligned p_z orbitals in formamide.

Stage 1 of 3

Comparison of Site A direct amide where continuous p_z orbitals enable delocalisation with Site B separated amine where intervening sp³ CH₂ group blocks conjugation.

The key step

Amide A has the supplied direct donor/carbonyl path. Saturated-spacer amine B lacks that ordinary contiguous path. Spatial proximity alone is insufficient, and

absence of this path is not an absolute zero-interaction theorem.

Formula reference • meaning and conditions

—

Equations and validity

Equation	What it tells you	When you use it
Check an ordinary contiguous appropriately aligned p-bearing path; this is a bounded model criterion.	Amide A has the supplied direct donor/carbonyl path. Saturated-spacer amine B lacks that ordinary contiguous path. Spatial proximity alone is insufficient, and absence of this path is not an absolute zero-interaction theorem.	• All at the conjugate network must unhybrid p orbitals • Adjacent orbitals have coplanar symmetry alignment

Worked explanation

Would folding the saturated-spacer molecule closer create the same continuous p pathway as removing the spacer?

Follow the operation, its justification and its result.

1. Local count and a continuous aligned pathway are independent checks.
2. No; spatial folding and orbital-connected adjacency are different changes.
3. Amide A has the supplied direct donor/carbonyl path. Saturated-spacer amine B lacks that ordinary contiguous path. Spatial proximity alone is

insufficient, and absence of this path is not an absolute zero-interaction theorem.

Result

No; spatial folding and orbital-connected adjacency are different changes.

Check

Changed case: replace a saturated interruption by a suitably aligned unsaturated bridge; reassess connectivity and orbital availability rather than reusing the answer.

Mistake clinic • diagnose and repair

A tempting wrong path

- Spatial proximity equals an orbital-connected p pathway.
- Recalling a local rule proves its applicability.
- A local count and label explain every geometry and reactivity difference.

Diagnose

- Would folding the saturated-spacer molecule closer create the same continuous p pathway as removing the spacer?
- Both sites have three sigma bonds and a lone pair. Is that alone sufficient to choose identical localized orbital-role descriptions?
- Would one bond/lone-pair count plus a label explain NH₃ shape, allene terminal-plane relation and amide donation by the same mechanism?

Repair

- Amide A has the supplied direct donor/carbonyl path. Saturated-spacer amine B lacks that ordinary contiguous path. Spatial proximity alone is insufficient, and absence of this path is not an absolute zero-interaction theorem.
- Four-domain counting is useful for localized introductory classification. Adjacent pi acceptors, aligned donor orbitals, planarity/donation evidence or radical occupancy can require a richer model.
- Use local count as a bounded starting model, then check residual p directions, occupancy, adjacency/alignment and evidence. A label alone

explains neither every geometry nor all reactivity. NH_3 , allene and amide stress different limitations.

1-2-3 QUICK CHECK

1 • CHECK

Changed case: replace a saturated interruption by a suitably aligned unsaturated bridge; reassess connectivity and orbital availability rather than reusing the answer.

2 • APPLY

Specify a counterfactual bridge and test all procedure conditions again.

3 • CONNECT

Local count and a continuous aligned pathway are independent checks.

2

Distinguish recalling a rule from establishing its applicability.

Hard concept

Section 9 of 10

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- Q6 · D3

The move learners miss: Inspect the supplied adjacency condition.

[Try it in Core2](#)

1 Recall the local rule with its scope.

Why valid: Four domains classify ordinary localized introductory centres; recalling it does not show that locality holds.

Result: Starting ledger: three N bonds plus one lone pair.

2 The step the hard question turns on Inspect the supplied adjacency condition.

Why valid: A has a carbonyl acceptor directly adjacent; B has a saturated

interruption in the ordinary p path.

Result: A needs a delocalisation/alignment test; B supports the local starting model here.

3 Identify evidence that defeats automatic application.

Why valid: Suitable donor/acceptor alignment, donation evidence or unusual occupancy add jobs the bond/pair total does not resolve.

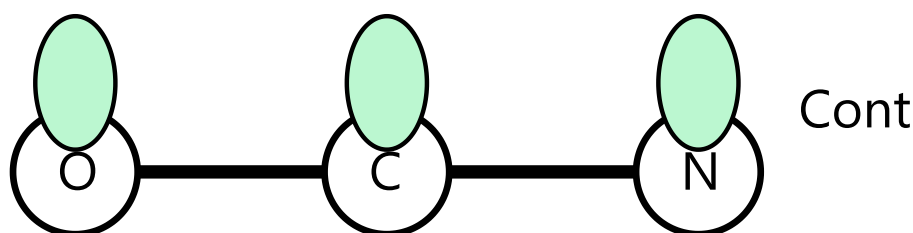
Result: Do not infer adequacy from the number four alone.

4 Separate known conditions from unknown quantities.

Why valid: Neither exact donor strength nor unique hybrid coefficients are supplied by this ledger.

Result: Apply a conditional classification or request the missing electronic evidence.

Site A · Direct amide: continuous aligned p-orbitals



Shows adjacent sp^2 centers with aligned p_z orbitals in formamide.

Stage 1 of 3

Comparison of Site A direct amide where continuous p_z orbitals enable delocalisation with Site B separated amine where intervening sp^3 CH₂ group blocks conjugation.

Worked explanation

Both sites have three sigma bonds and a lone pair. Is that alone sufficient to choose identical localized orbital-role descriptions?

Follow the operation, its justification and its result.

1. Local counting assumes the localized/domain model is useful for the asked task.

2. No; supplied adjacency/alignment evidence differs.
3. Four-domain counting is useful for localized introductory classification. Adjacent pi acceptors, aligned donor orbitals, planarity/donation evidence or radical occupancy can require a richer model.

Result

No; supplied adjacency/alignment evidence differs.

Check

Changed check: the same formal lone-pair entry at an amine and an amide can have different orbital-role descriptions under the supplied conditions.

Mistake clinic • diagnose and repair

—

A tempting wrong path

- Spatial proximity equals an orbital-connected p pathway.
- Recalling a local rule proves its applicability.
- A local count and label explain every geometry and reactivity difference.

Diagnose

- Would folding the saturated-spacer molecule closer create the same continuous p pathway as removing the spacer?
- Both sites have three sigma bonds and a lone pair. Is that alone sufficient to choose identical localized orbital-role descriptions?
- Would one bond/lone-pair count plus a label explain NH₃ shape, allene terminal-plane relation and amide donation by the same mechanism?

Repair

- Amide A has the supplied direct donor/carbonyl path. Saturated-spacer amine B lacks that ordinary contiguous path. Spatial proximity alone is insufficient, and absence of this path is not an absolute zero-interaction theorem.
- Four-domain counting is useful for localized introductory classification. Adjacent pi acceptors, aligned donor orbitals, planarity/donation evidence or radical occupancy can require a richer model.

- Use local count as a bounded starting model, then check residual p directions, occupancy, adjacency/alignment and evidence. A label alone explains neither every geometry nor all reactivity. NH₃, allene and amide stress different limitations.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: the same formal lone-pair entry at an amine and an amide can have different orbital-role descriptions under the supplied conditions.

2 • APPLY

Separate a recalled count from a model-selection decision in a new conjugated case.

3 • CONNECT

Local counting assumes the localized/domain model is useful for the asked task.

3

Audit a proposed universal rule across cases that stress different boundaries.

Hard concept

Section 10 of 10

← Previous

Why this section matters for the question set

This unit builds toward

- Q10 • D4

The move learners miss: Retain local counting as a starting ledger.

[Try it in Core2](#)

1 Audit each case for the explanatory job omitted by the slogan.

Why valid: NH₃ needs lone-pair versus nuclear-shape distinction; allene needs residual-p orientation; formamide needs donation conditions; the separated amine needs a pathway test.

Result: A label alone cannot supply all four explanations.

2 The step the hard question turns on Retain local counting as a starting ledger.

Why valid: Atom-local sigma directions and central pairs give useful introductory information when localized assumptions hold.

Result: Count locally and conserve the valence basis.

3 Add occupancy, adjacency and alignment branches.

Why valid: A residual p function must have a specified occupancy and suitable neighbouring orbital path for the proposed interaction.

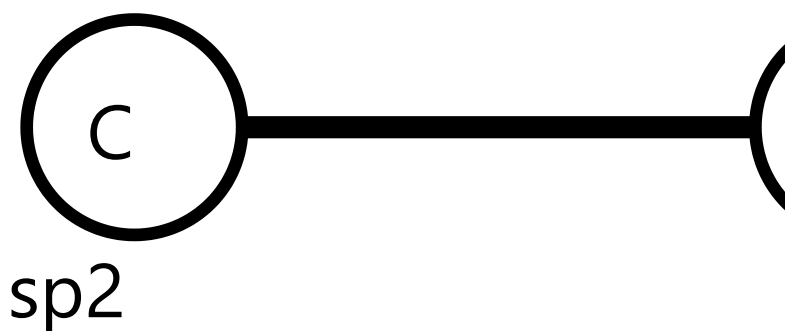
Result: Check what orbitals do, rather than treating the hybrid label as their cause.

4 Include a validity and evidence stopping rule.

Why valid: Quantitative geometry/reactivity or competing compatible electronic accounts require evidence beyond the ledger.

Result: Use a conditional framework and seek an explicit model or measurement when the simple rule is inadequate.

1 • Linear sigma axis and orbital hy



Shows linear C–C–C axis with sp^2 – sp – sp^2 hybridization.

Stage 1 of 4

Staged orbital model showing the central sp carbon of allene having two orthogonal p orbitals forming independent π bonds with terminal sp^2 carbons,

resulting in mutually perpendicular terminal CH₂ planes.

Worked explanation

Would one bond/lone-pair count plus a label explain NH₃ shape, allene terminal-plane relation and amide donation by the same mechanism?

Follow the operation, its justification and its result.

1. Counts, orbital roles, geometry and physical evidence are distinct layers.
2. No; their explanatory jobs differ. Repeating label-as-final-cause across all cases exposes an overgeneralised framework.
3. Use local count as a bounded starting model, then check residual p directions, occupancy, adjacency/alignment and evidence. A label alone explains neither every geometry nor all reactivity. NH₃, allene and amide stress different limitations.

Result

No; their explanatory jobs differ. Repeating label-as-final-cause across all cases exposes an overgeneralised framework.

Check

Changed check: apply the framework to CH₃ radical; three bonds plus one unpaired electron require an occupancy step absent from the original slogan.

Mistake clinic • diagnose and repair

—

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1-2-3 QUICK CHECK

1 • CHECK

Changed check: apply the framework to CH₃ radical; three bonds plus one unpaired electron require an occupancy step absent from the original slogan.

2 • APPLY

Produce a qualified procedure with an explicit stop-and-look-for-more-evidence branch.

3 • CONNECT

Counts, orbital roles, geometry and physical evidence are distinct layers.

Question-specific diagnosis and repair

—

Repair for Q4

Test the rule on a changed case

Would folding the saturated-spacer molecule closer create the same continuous p pathway as removing the spacer?

Your prediction and reason

Repair for Q6

Test the rule on a changed case

Both sites have three sigma bonds and a lone pair. Is that alone sufficient to choose identical localized orbital-role descriptions?

Your prediction and reason

Repair for Q10

Test the rule on a changed case

Would one bond/lone-pair count plus a label explain NH₃ shape, allene terminal-plane relation and amide donation by the same mechanism?

Your prediction and reason

Now you do one

An introductory local model proposes a direct aligned donor/acceptor path. Insert a saturated spacer. State the local-model effect and one claim it does not establish.

Model answer

The ordinary contiguous p pathway is interrupted. This does not prove zero interaction of every kind or predict a numerical basicity/energy change.

1. The ordinary contiguous p pathway is interrupted. This does not prove zero interaction of every kind or predict a numerical basicity/energy change.

Practice this concept in Core2

Q4

Q6

Q10